



ROB 2040 Studio 5000 LABS



OAKLAND COMMUNITY COLLEGE®

ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

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LAB: WASHER STATION

PROCESS FILE: WASHER

OBJECTIVE: Create ladder structure and assign data tags.

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

Create a program that when a momentary normally open pushbutton is pressed and held, two clamp solenoids will be energized. A normally open limit switch for each clamp will be set when that clamp is in position. When the two clamps are in position, a spray wash will clean the part. Releasing the pushbutton will turn off the spray wash and clamps.

I/O CONFIGURATION: (Processor and I/O has been configured)

Slot 2 - Studio Emulate 5000 Controller

Slot 3 - 1756-IB16

Slot 4 - 1756

NOTE: All Inputs and Outputs are BOOLEAN Data Type shown in a DECIMAL Style.

INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.0	PUSHBUTTON	Pushbutton Wash Cycle
Local:3:I.Data.1	SWITCH_CLAMP1	Limit Switch Clamp 1
Local:3:I.Data.2	SWITCH_CLAMP2	Limit Switch Clamp 2

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	SOLENOID_CLAMP1	Clamp 1 Solenoid
Local:4:O.Data.1	SOLENOID_CLAMP2	Clamp 2 Solenoid
Local:4:O.Data.2	WASH_VALVE	Spray Wash Valve

Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution "Change processor mode to Run?", select **Yes**.

HMI:

- ① Operator Pressing and Holding the START pushbutton:

Placing the cursor over the **Start** pushbutton, left click and hold the mouse button.

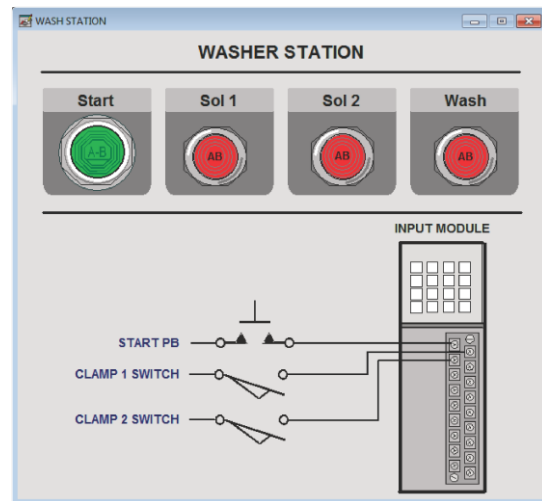
If you do not hold the left mouse button the pushbutton is released.

- ② The two outputs for the solenoid valves for the clamps turns ON. The **Sol1** and **Sol2** indicators will illuminate green.

- ③ After the input switches for the **Clamp 1** and **Clamp2** are set, the spray **Wash** valve turns ON and indicator will illuminate green.

- ④ To turn off the washer, release the left mouse button. This releases the **Start** pushbutton.

- ⑤ The spray wash and the solenoid valves turns OFF and the inputs for the clamps reset.



Offline:

To return to Offline

- At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- At the caution "Change processor mode to Remote Program?", select **Yes**.
- At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: CYLINDER INTERLOCKS

PROCESS FILE: CYLINDER INTERLOCKS

OBJECTIVE: Illustrate the use of Examine Input Opened instructions as interlocks.
using input and output addresses.

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

The three circuits use Examine If Opened (XIO) instruction to prevent a solenoid from being energized in both the forward and reverse directions. See: CYLINDER INTERLOCK PROGRAM.

CIRCUIT 1:

The interlocking XIO instructions for the output solenoids are addressed to the opposing controlling input address. This will result in both solenoids de-energizing if the cylinder is commanded for both forward and reverse directions.

CIRCUIT 2:

The interlocking XIO instructions for the output solenoids are addressed to the opposing solenoid output address. This will prevent the opposing solenoids from energizing if the cylinder is commanded for both forward and reverse directions.

CIRCUIT 3:

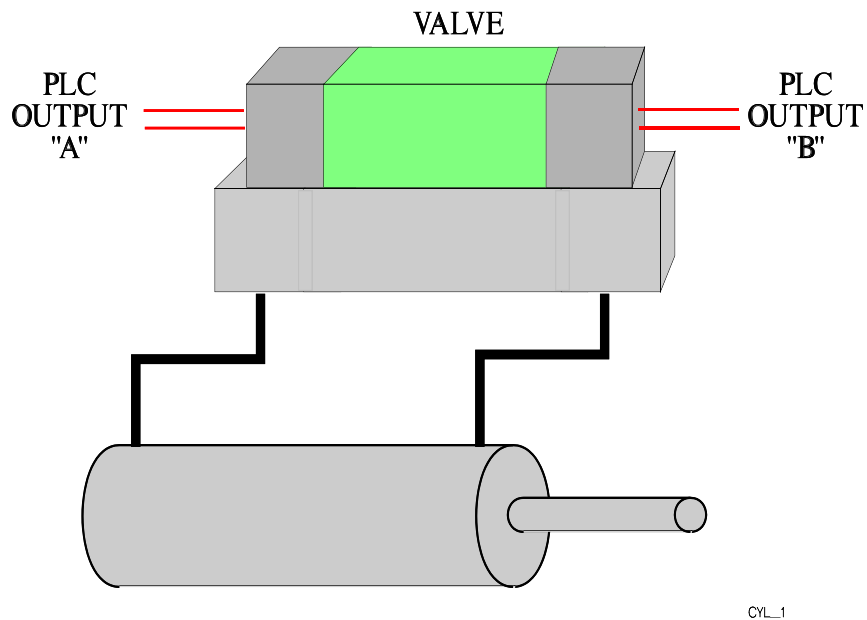
The XIO instruction addressed to the reverse control is used in the rung for the forward solenoid and will override the forward command. If currently commanded to go forward, then the reverse command is given also - the cylinder will go into reverse.

I/O CONFIGURATION: (Processor and I/O has been configured)

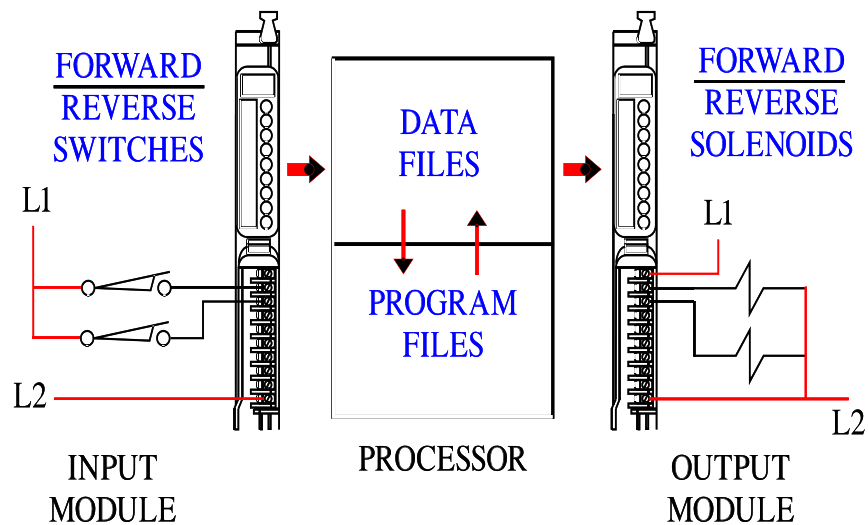
Slot 2 - Studio Emulate 5000 Controller
Slot 3 - 1756-IB16
Slot 4 - 1756

[See Page 5 and 6 for the Tag assignments and Program logic.](#)

CYLINDER INTERLOCKS



CYLINDER INTERLOCKS



CYL 2

I/O CONFIGURATION: (Processor and I/O has been configured)

Slot 2 - Studio Emulate 5000 Controller

Slot 3 - 1756-IB16

Slot 4 - 1756

NOTE: All Inputs and Outputs are BOOLEAN Data Type shown in a DECIMAL Style.

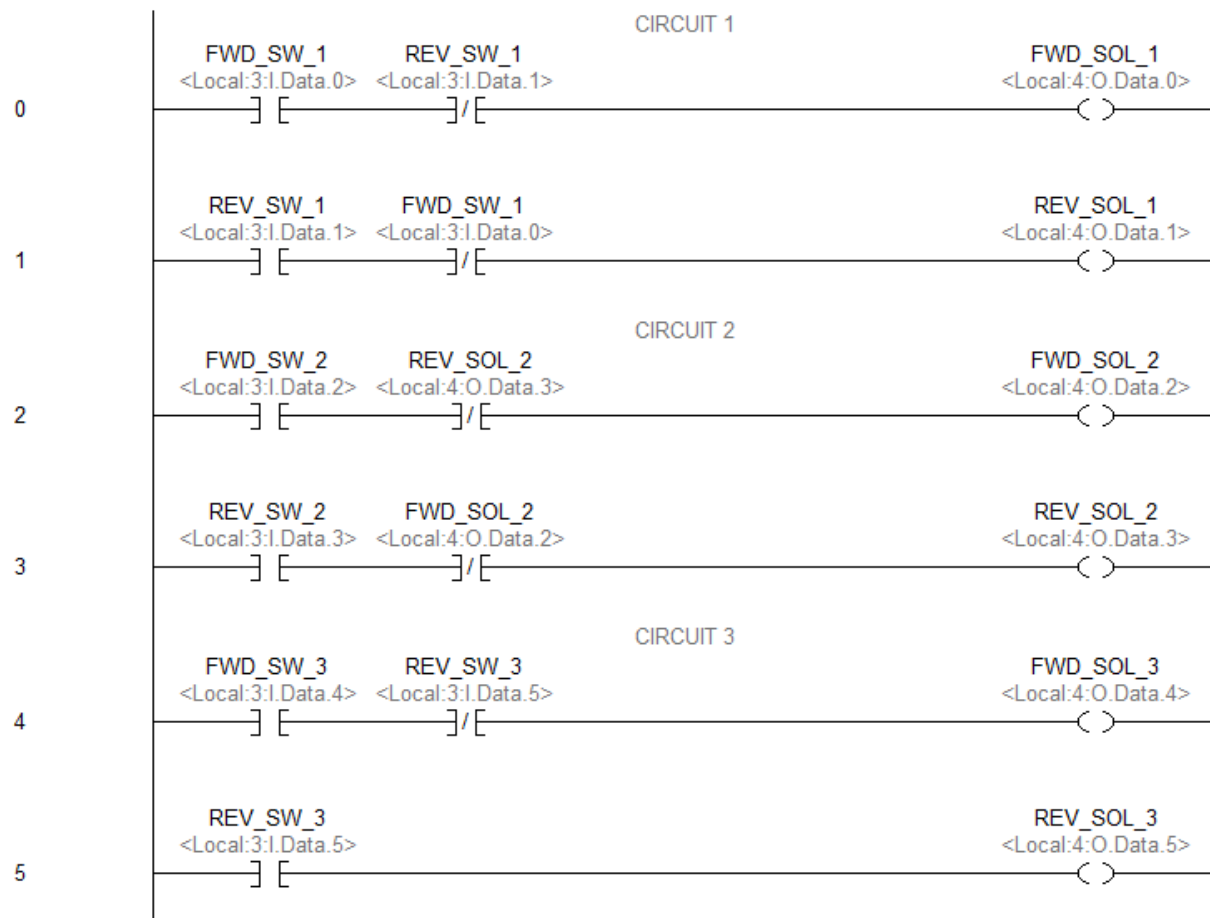
INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.0	FWD_SW_1	Forward Switch 1
Local:3:I.Data.1	REV_SW_1	Reverse Switch 1
Local:3:I.Data.2	FWD_SW_2	Forward Switch 2
Local:3:I.Data.3	REV_SW_2	Reverse Switch 2
Local:3:I.Data.4	FWD_SW_3	Forward Switch 3
Local:3:I.Data.5	REV_SW_3	Reverse Switch 3

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	FWD_SOL_1	Forward Solenoid 1
Local:4:O.Data.1	REV_SOL_1	Reverse Solenoid 1
Local:4:O.Data.2	FWD_SOL_2	Forward Solenoid 2
Local:4:O.Data.3	REV_SOL_2	Reverse Solenoid 2
Local:4:O.Data.4	FWD_SOL_3	Forward Solenoid 3
Local:4:O.Data.5	REV_SOL_3	Reverse Solenoid 3

Cylinders Program



Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution “Change processor mode to Run?”, select **Yes**.

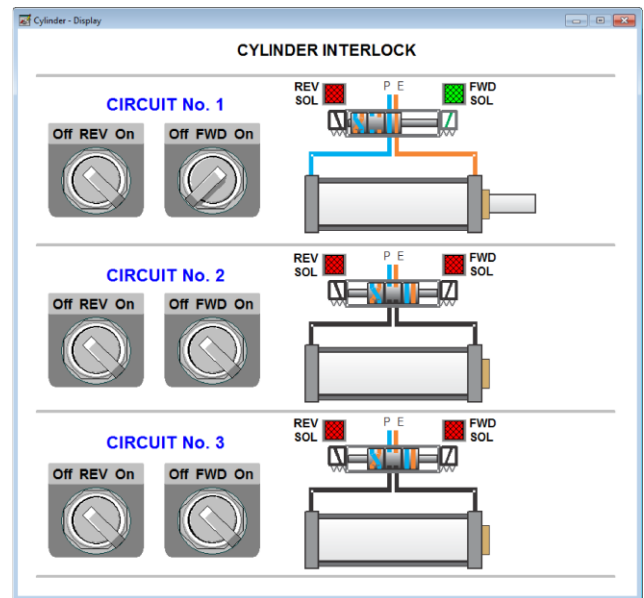
Notes on Running:



This lab and all following labs will have an HMI, use the document: **FactoryTalk View SE Client** for the procedures to open and start the HMI.

Each circuit has a Forward and Reverse switch to control the direction of the cylinder.

Use the switches to observe the interlocked actions when a cylinder is commanded to go Forward and Reverse at the same time.



Offline:

To return to Offline

- At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- At the caution "Change processor mode to Remote Program?", select **Yes**.
- At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: SS1**PROCESS FILE:** SS1**OBJECTIVE:** Using start-stop button with a holding circuit. Use of Normally Closed pushbutton.**SOFTWARE:** Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

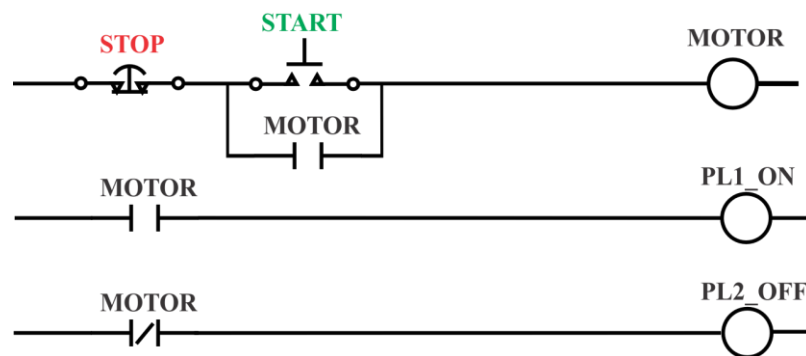
Create the control of a START - STOP circuit for a single motor control with ON - OFF indicators.

I/O CONFIGURATION: (Processor and I/O has been configured)

Slot 2 - Studio Emulate 5000 Controller

Slot 3- 1756-IB16

Slot 4 - 1756-OB16E

STOP-START CIRCUIT WITH INDICATOR**INPUT:**

Alias Address	Controller Tag	Description
Local:3:I.Data.0	STOP	Momentary NC Pushbutton Stop
Local:3:I.Data.1	START	Momentary NO Pushbutton Start

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	MOTOR	Motor Control Relay
Local:4:O.Data.1	PL1_ON	Indicator Light ON
Local:4:O.Data.2	PL2_OFF	Indicator Light OFF

Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution “Change processor mode to Run?”, select **Yes**.

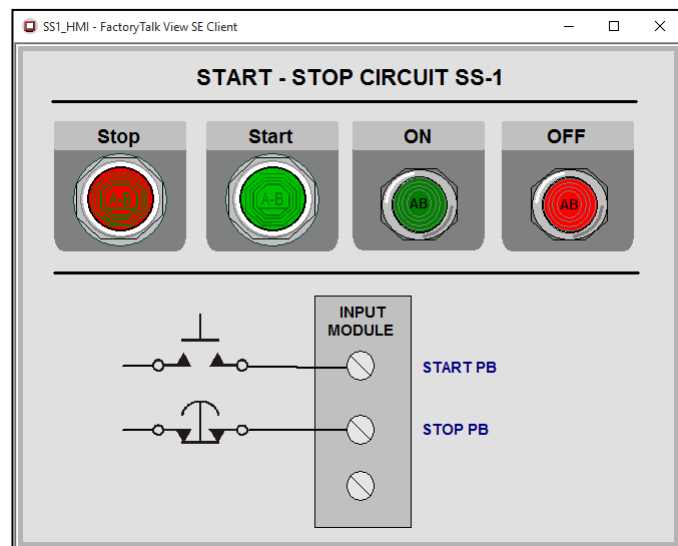
HMI:

- ① Operator Pressing the START:
Left click on the **Start** pushbutton.

ON indicator illuminated.

- ② Operator Pressing the STOP:
Left click on the **Stop** pushbutton.

OFF indicator illuminated.



Offline:

To return to Offline

- At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- At the caution “Change processor mode to Remote Program?”, select **Yes**.
- At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: SS2**PROCESS FILE:** SS2**OBJECTIVE:** Using selector switch with a holding circuit for a JOG/RUN circuit.**SOFTWARE:** Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

Create the control of a START - STOP circuit that will allow for the continuous running of a motor control, or by selecting the momentary (jogging) motion of the controlled motor when the START pushbutton is pressed.

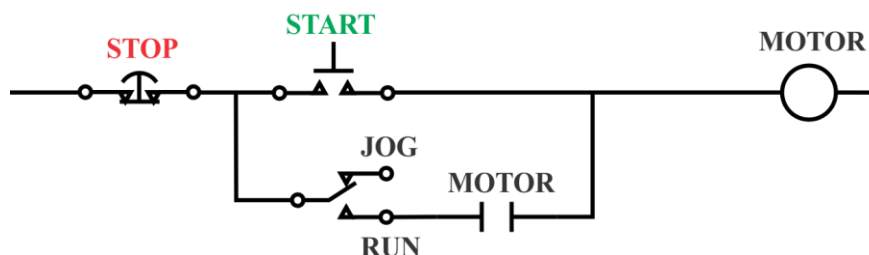
NOTE: The selector switch for JOG - RUN is a two-position maintained contact selector switch. Only the RUN contact of the switch is wired to the terminal on the PLC module.

I/O CONFIGURATION: (Processor and I/O has been configured)

Slot 2 - Studio Emulate 5000 Controller

Slot 3- 1756-IB16

Slot 4 - 1756-OB16E

JOG/RUN SELECTOR SWITCH**INPUT:**

Alias Address	Controller Tag	Description
Local:3:I.Data.0	STOP	Momentary NC Pushbutton Stop
Local:3:I.Data.1	START	Momentary NO Pushbutton Start
Local:3:I.Data.2	SEL_SW1	Selector Switch for JOG or RUN

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	MOTOR	Motor Control Relay

Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- a. At the ONLINE BAR go from **OFFLINE** to **Download**.
- b. At the caution, select the **Download** button.
- c. At the ONLINE BAR go from **Rem Prog** to **Run**.
- d. At the caution "Change processor mode to Run?", select **Yes**.

HMI:

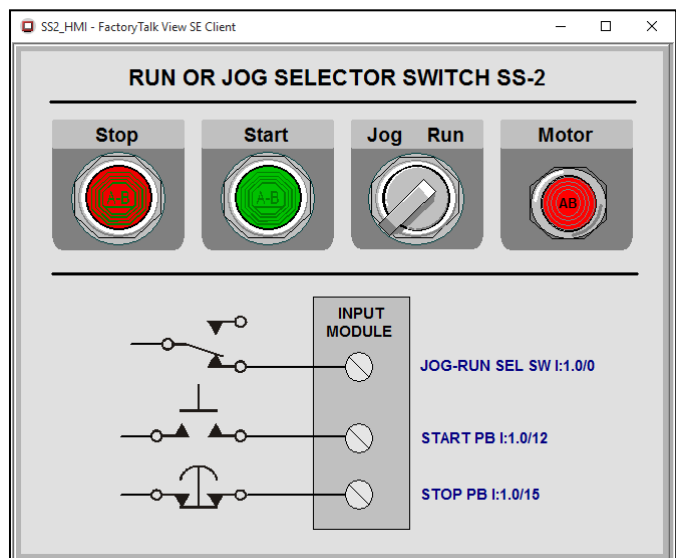
To RUN the motor:

- ① Left click on the **Jog/Run** switch so the switch is at the Run position.
- ② Left click on the **Start** pushbutton.

MOTOR indicator is illuminated green.

- ③ Left click on the **Stop** pushbutton.

MOTOR indicator is illuminated red.



To JOG the motor:

- ① Left click on the **Jog/Run** switch so the switch is at the Jog position.
- ② Left click and hold mouse button over the **Start** pushbutton.

MOTOR indicator is illuminated green.

- ③ Release the left mouse button over the **Start** pushbutton.

MOTOR indicator is illuminated red

Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: SS3**PROCESS FILE:** SS3**OBJECTIVE:** Using multiple start-stop button with a holding circuit.**SOFTWARE:** Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

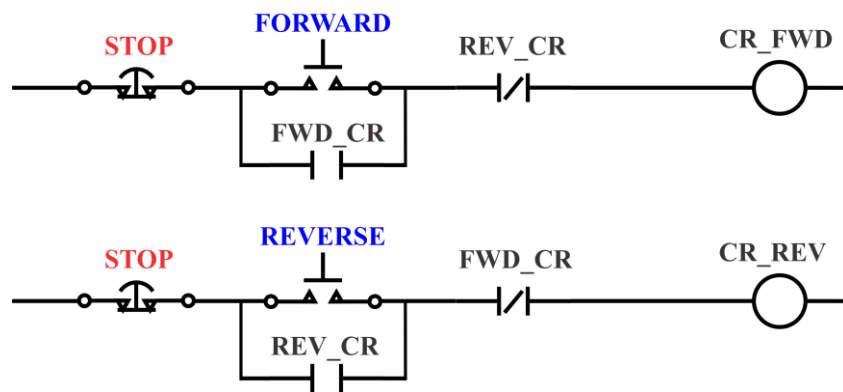
Create the control of a single motor with FORWARD and REVERSE control relays where the motor must be stopped before changing direction.

I/O CONFIGURATION: (Processor and I/O has been configured)

Slot 2 - Studio Emulate 5000 Controller

Slot 3- 1756-IB16

Slot 4 - 1756-OB16E

FORWARD/REVERSE WITH STOPPING**INPUT:**

Alias Address	Controller Tag	Description
Local:3:I.Data.0	STOP	Momentary NC Pushbutton Stop
Local:3:I.Data.1	FWD	Momentary NO Pushbutton Forward
Local:3:I.Data.2	REV	Momentary NO Pushbutton Reverse

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	CR_FWD	Motor Control Relay Forward
Local:4:O.Data.1	CR_REV	Motor Control Relay Reverse

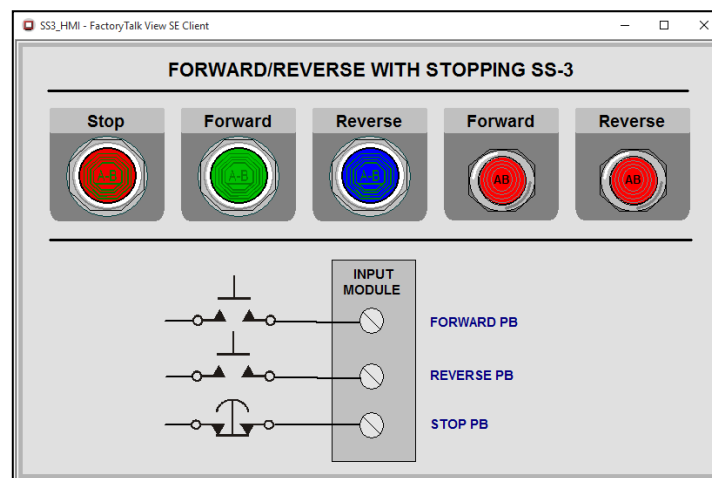
Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- a. At the ONLINE BAR go from **OFFLINE** to **Download**.
- b. At the caution, select the **Download** button.
- c. At the ONLINE BAR go from **Rem Prog** to **Run**.
- d. At the caution "Change processor mode to Run?", select **Yes**.

HMI:



- ① Operator Pressing the FORWARD:
Left click on the **Forward** pushbutton.

The **Forward** indicator is illuminated green.

- ② Operator Pressing the REVERSE:
Left click on the **Reverse** pushbutton.

No change, direction cannot be changed without stopping first.

- ③ Operator Pressing the STOP:
Left click on the **Stop** pushbutton.
Both motor indicators are illuminated red.

- ④ Operator Pressing the REVERSE:
Left click on the **Reverse** pushbutton.

The **Reverse** indicator is illuminated green.

⑤ Operator Pressing the FORWARD:

Left click on the **Forward** pushbutton.

No change, direction cannot be changed without stopping first.

⑥ Operator Pressing the STOP:

Left click on the **Stop** pushbutton.

Both motor indicators are illuminated red.

Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: CONVEYOR**PROCESS FILE:** CONVEYOR**OBJECTIVE:** Programming normally closed field devices.**SOFTWARE:** Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE**NOTE:** Open program and save as with new name.

I/O CONFIGURATION: (Processor and I/O has been configured)

Slot 2 - Studio Emulate 5000 Controller

Slot 3 - 1756-IB16

Slot 4 - 1756OB16E

Create a program for a conveyor system to transport a single parts container. When a bin is placed on the line, the operator will energize the conveyor via a Start/Stop pushbutton panel. When the bin reaches the end of the line, a photoelectric switch de-energizes the conveyor.

The photoelectric switch is wired normally closed and any absence of the signal, that is the beam is interrupted, will stop the conveyor.

When creating the tag, the description MUST BE entered.

INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.0	START	Momentary Normally Open Pushbutton
Local:3:I.Data.1	STOP	Momentary Normally Closed Pushbutton
Local:3:I.Data.2	PE1	Photoelectric Switch Normally Closed

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	MCR1	Motor Control Relay for Conveyor Motor
Local:4:O.Data.1	PL1	Pilot Light (GREEN) Conveyor ON
Local:4:O.Data.2	PL2	Pilot Light (RED) Conveyor OFF

Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution “Change processor mode to Run?”, select **Yes**.

HMI:

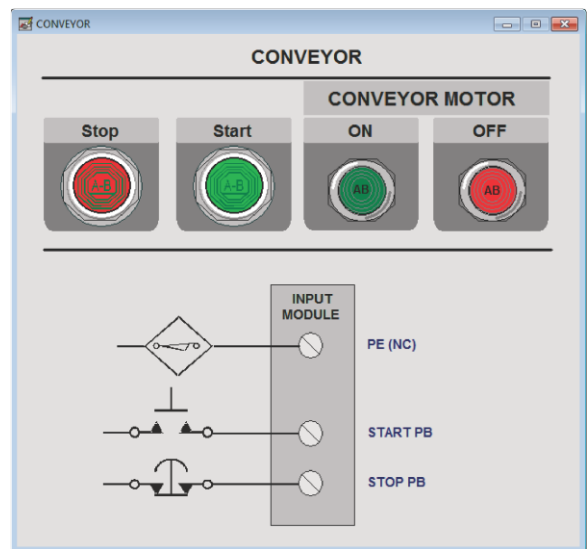
① Operator Pressing the START pushbutton:

Left click on the **Start** pushbutton.

The **ON** motor indicator is illuminated.

② The interrupting of the photoelectric at the end of the conveyor to stop the motor is simulated by the emulation environment.

The **OFF** motor indicator is illuminated.



Offline:

To return to Offline

- At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- At the caution “Change processor mode to Remote Program?”, select **Yes**.
- At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: MIXING VAT**PROCESS FILE:** VAT**OBJECTIVE:** Use of internal to control sequence of operations and scope of tags.**SOFTWARE:** Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE**NOTE:** Open program and save as with new name.

When a START pushbutton is pressed/released the filler valve on the vat will open and the drain valve will close. A normally closed float switch will open when the tank is filled, and the filler valve will be shutoff. The motor will then start to mix the contents of the tank.

The pressing/releasing of the STOP pushbutton will turn off the motor and open the drain valve.

NOTE: The pressing of START pushbutton closes the de-energizes the drain valve.
When de-energized by the logic, the Filler valve is closed.

I/O CONFIGURATION: (Processor and I/O has been configured)

Slot 2 - Studio Emulate 5000 Controller

Slot 3- 1756-IB16

Slot 4 - 1756-OB16E

NOTE: All Inputs, Outputs, and Internal are BOOLEAN Data Type shown in a DECIMAL Style.

When creating the tag, the description MUST BE entered.

INPUT: Note: inputs use Program Parameter and Local Tags.

Alias Address	Param/Local Tag	Description
Local:3:I.Data.6	START	START Pushbutton Momentary N. O.
Local:3:I.Data.7	STOP	STOP Pushbutton Momentary N. C.
Local:3:I.Data.0	FLOAT_SWITCH	Vat Filled Float Switch N. C.

OUTPUT: [Note: outputs use Controller Tags.](#)

Alias Address	Controller Tag	Description
Local:4:O.Data.0	DRAIN_VALVE	Vat Drain Valve
Local:4:O.Data.1	FILLER_VALVE	Vat Filler Valve
Local:4:O.Data.2	MIX_MOTOR	Vat Motor Control

INTERNAL: [Note: the internal is a Program Parameter and Local Tag.](#)

Param/Local Tag	Data Type	Description
INTERNAL	Boolean	Enable 1- Fill/Mix Cycle 0 - Drain Cycle

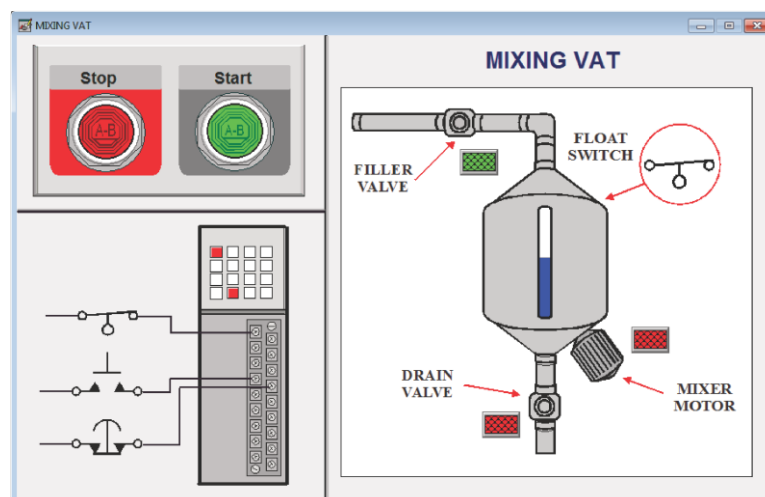
Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution "Change processor mode to Run?", select **Yes**.

HMI:



- ① Operator Pressing the START pushbutton:
Left click on the **Start** pushbutton.
- ② The **DRAIN VALVE** turns OFF and the **FILLER VALVE** will turn ON.
- ③ When the **NORMALLY CLOSED FLOAT SWITCH** breaks, the **FILLER VALVE** will turn OFF and the **MIXER MOTOR** will turn ON.
- ④ Operator Pressing the STOP pushbutton:
Left click on the **Stop** pushbutton.

The **MIXER MOTOR** will turn OFF and the **DRAIN VALVE** turns ON.

Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: TIMER ON STATUS BITS

PROCESS FILE: TIMER ON

OBJECTIVE: Program a TIMER ON instruction and the status bits.

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

Create the program and complete the following questions. All inputs are XIC instructions.

When creating the tag, the description MUST BE entered.

INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.1	SWITCH	Timer Control N. O. Switch

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	ENABLE	ENABLE Status Indicator
Local:4:O.Data.1	TIMER_TIMING	TIMER TIMING Status Indicator
Local:4:O.Data.2	DONE	DONE Status Indicator

INTERNAL:

Param/Local Tag	Data Type	Description
DWELL_ON	TIMER	Timer On
DWELL_ON.EN ⁽¹⁾		ENABLE Status Bit
DWELL_ON.TT ⁽¹⁾		TIMER TIMING Status Bit
DWELL_ON.DN ⁽¹⁾		DONE Status Bit

(1)Timer status bits are created automatically.

Studio 5000 Logix Emulate must be started before Downloading

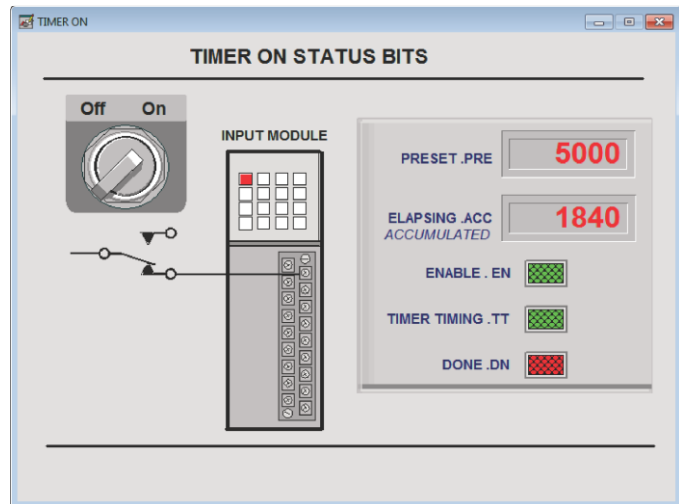
Download:

Download the project and Run.

- a. At the ONLINE BAR go from **OFFLINE** to **Download**.
- b. At the caution, select the **Download** button.
- c. At the ONLINE BAR go from **Rem Prog** to **Run**.
- d. At the caution "Change processor mode to Run?", select **Yes**.

HMI:

- ① Turn switch to OFF.
- ② Turn switch to ON.
- ③ When PRESET and ACCUMULATED are equal turn switch to OFF.
- ④ Turn switch to ON.
- ⑤ While ACCUMULATED is elapsing, turn switch to OFF.



Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: TIMER OFF STATUS BITS

PROCESS FILE: TIMER OFF

OBJECTIVE: Program a TIMER OFF instruction and the status bits.

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

Create the program and complete the following questions. All inputs are XIC instructions.

When creating the tag, the description MUST BE entered.

INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.10	SWITCH	Timer Control N. O. Switch

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.10	ENABLE	ENABLE Status Indicator
Local:4:O.Data.11	TIMER_TIMING	TIMER TIMING Status Indicator
Local:4:O.Data.12	DONE	DONE Status Indicator

INTERNAL:

Param/Local Tag	Data Type	Description
DWELL_OFF	TIMER	Timer Off
DWELL_OFF.EN ⁽¹⁾		ENABLE Status Bit
DWELL_OFF.TT ⁽¹⁾		TIMER TIMING Status Bit
DWELL_OFF.DN ⁽¹⁾		DONE Status Bit

(1)Timer status bits are created automatically.

Studio 5000 Logix Emulate must be started before Downloading

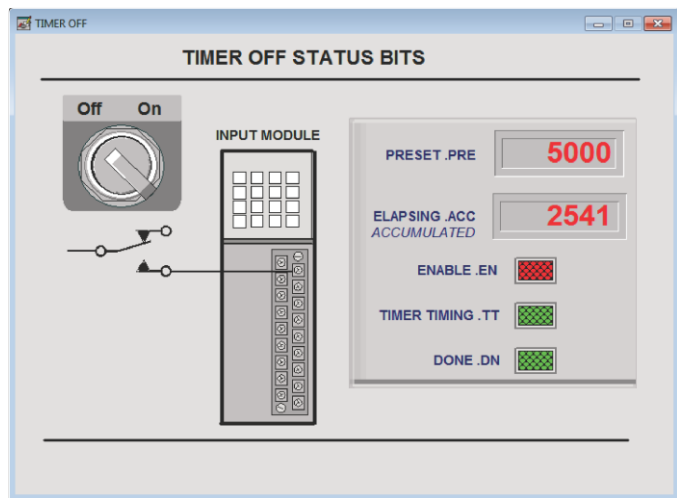
Download:

Download the project and Run.

- a. At the ONLINE BAR go from **OFFLINE** to **Download**.
- b. At the caution, select the **Download** button.
- c. At the ONLINE BAR go from **Rem Prog** to **Run**.
- d. At the caution "Change processor mode to Run?", select **Yes**.

HMI:

- ① Turn switch to ON.
- ② Turn switch to OFF.
- ③ When PRESET and ACCUMULATED are equal turn switch to ON.
- ④ Turn switch to OFF.
- ⑤ While ACCUMULATED is elapsing, turn switch to ON.



Offline:

1. To return to Offline
 - a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
 - b. At the caution "Change processor mode to Remote Program?", select **Yes**.
 - c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: STARTUP TIMER

PROCESS FILE: STARTUP

OBJECTIVE: Determine the type of timer required for the required logic and program the appropriate relay instruction controlled by the DONE bit.

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

When a START pushbutton is pressed/released, a warning horn will sound for five (5) seconds. When the horn turns off, an output will then be turned on to run a conveyor. The conveyor can be turned off by the STOP pushbutton.

When creating the tag, the description MUST BE entered.

INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.0	START	Momentary N.O. START Pushbutton
Local:3:I.Data.1	STOP	Momentary N.C. STOP Pushbutton

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	HORN	Warning Horn
Local:4:O.Data.1	MOTOR	Conveyor Motor

INTERNAL: Notice **WARN_DWELL** is a Controller Tag

Controller Tag	Data Type	Description
WARN_DWELL	TIMER	Start-up Timer
WARN_DWELL.DN ⁽¹⁾		Timer Done Status Bit
WARN_DWELL.EN ⁽¹⁾		Timer Enable Status Bit

⁽¹⁾ Timer status bits created automatically.

Optional

WARN_DWELL.TT ⁽¹⁾		Timer Timing Status Bit
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⁽¹⁾ Timer status bits are created automatically.

Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- a. At the ONLINE BAR go from **OFFLINE** to **Download**.
- b. At the caution, select the **Download** button.
- c. At the ONLINE BAR go from **Rem Prog** to **Run**.
- d. At the caution "Change processor mode to Run?", select **Yes**.

HMI:

① Operator Pressing the START pushbutton:

Left click on the **Start** pushbutton.

② The **HORN** will sound.

③ After five seconds the **HORN** turn OFF and the conveyor **MOTOR** turns ON with the indicator illuminated green.



④ Operator Pressing the STOP pushbutton:

Left click on the **Stop** pushbutton.

The **MOTOR** indicator is illuminated red.

Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: TIMED WASHER STATION

PROCESS FILE: WASH TON

OBJECTIVE: Identify the type of timer instruction and the controlling logic.

HMI: WASH_HMI

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

Modify the WASHER STATION lab so that a timer will allow the spray wash to be on for 10 sec. while the pushbutton is being held.

When creating the tag, the description MUST BE entered.

INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.0	PUSHBUTTON	Pushbutton Wash Cycle
Local:3:I.Data.1	SWITCH_CLAMP1	Limit Switch Clamp 1
Local:3:I.Data.2	SWITCH_CLAMP2	Limit Switch Clamp 2

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	SOLENOID_CLAMP1	Clamp 1 Solenoid
Local:4:O.Data.1	SOLENOID_CLAMP2	Clamp 2 Solenoid
Local:4:O.Data.2	WASH_VALVE	Spray Wash Valve

INTERNAL:

Param/Local Tag	Data Type	Description
WASH_DELAY	TIMER	Spray Wash Timer
WASH_DELAY.DN ⁽¹⁾		Spray Wash Delay Complete

(1) Timer status bits are created automatically.

OPTIONAL:

WASH_DELAY.EN ⁽¹⁾		Spray Wash Delay Enabled
WASH_DELAY.TT ⁽¹⁾		Spray Wash Delay Timing

(1) Timer status bits are created automatically.

Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution "Change processor mode to Run?", select **Yes**.

HMI:

① Operator Pressing and Holding the START pushbutton:

Placing the cursor over the **Start** pushbutton, left click and hold the mouse button.

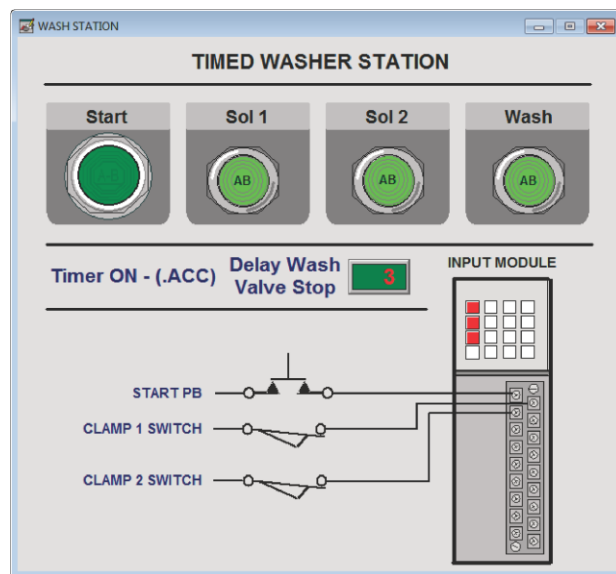
If you do not hold the left mouse button the pushbutton is released.

② The two outputs for the solenoid valves for the clamps turns ON. The **Sol1** and **Sol2** indicators will illuminate green.

③ After the input switches for the **Clamp 1** and **Clamp2** are set, the spray **Wash** valve turns ON and the indicator will illuminate green.

④ After ten seconds the spray **Wash** turns OFF.

⑤ To turn open the clamps, release the left mouse button. This releases the **Start** pushbutton and the solenoid valves turns OFF and the inputs for the clamps reset



Offline:

To return to Offline

- At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- At the caution "Change processor mode to Remote Program?", select **Yes**.
- At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: BUILDING LIGHTS

PROCESS FILE: BUILDING

OBJECTIVE: Use of TOF instruction and cascading timer logic.

HMI: BUILDING_HMI

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

Create a program that will use cascading timers to control when building and parking lot lights are turned off. When the light switch is switched to the off position, a Normally Open switch returned to the normal position, the building light will turn off immediately. The lights for the Exit Door will remain on for 5 seconds. The Parking Lot lights will remain on for an additional 10 seconds. *(Time values are for testing; actual application would use appropriate time values.)*

When creating the tag, the description MUST BE entered.

INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.0	LIGHT_SWITCH	Building Switch

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	BUILDING_LIGHTS	Building Lights
Local:4:O.Data.1	EXIT_LIGHTS	Exit Door Lights
Local:4:O.Data.2	LOT_LIGHTS	Parking Lot Lights

INTERNAL:

Param/Local Tag	Data Type	Description
EXIT_DELAY	TIMER	Timer for Exit
EXIT_DELAY.DN ⁽¹⁾		Exit Light Off
LOT_DELAY	TIMER	Timer for Parking Lot
LOT_DELAY.DN ⁽¹⁾		Parking Lights Off

(1)Timer status bits are created automatically.

Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- a. At the ONLINE BAR go from **OFFLINE** to **Download**.
- b. At the caution, select the **Download** button.
- c. At the ONLINE BAR go from **Rem Prog** to **Run**.
- d. At the caution "Change processor mode to Run?", select **Yes**.

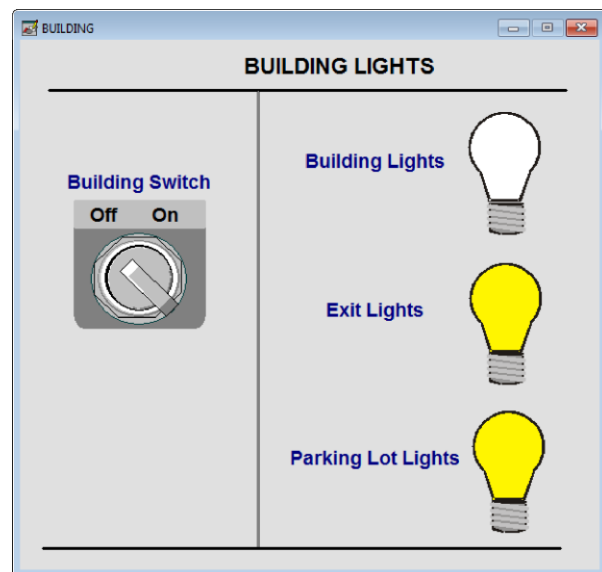
HMI:

① Turn Switch to ON.

- All the lights should turn on.

② Turn switch to OFF.

- The building lights will turn off immediately.
- The exit lights will turn off after 5 seconds.
- The parking lot lights will turn off after an additional 10 seconds.



Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: TIMED WASHER STATION - RETENTIVE TIMER

PROCESS FILE: WASH RTO

OBJECTIVE: Use of retentive timer during wash cycle and identifying logic to reset timer.

HMI: WASH_HMI

SOFTWARE: Studio500, RSEmulate500, and FactoryTalk SE

Use a **Retentive Timer On (RTO)** instruction for controlling the washer valve.

If the START pushbutton is released before the timing cycle is complete, the accumulated value of the timer *will not be reset*. Pressing the pushbutton will start the timing cycle from the current accumulated value.

When creating the tag, the description MUST BE entered.

INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.0	PUSHBUTTON	Pushbutton Wash Cycle
Local:3:I.Data.1	SWITCH_CLAMP1	Limit Switch Clamp 1
Local:3:I.Data.2	SWITCH_CLAMP2	Limit Switch Clamp 2

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	SOLENOID_CLAMP1	Clamp 1 Solenoid
Local:4:O.Data.1	SOLENOID_CLAMP2	Clamp 2 Solenoid
Local:4:O.Data.2	WASH_VALVE	Spray Wash Valve

INTERNAL:

Param/Local Tag	Data Type	Description
WASH_DELAY	TIMER	Spray Wash Timer
WASH_DELAY.DN ⁽¹⁾		Spray Wash Delay Complete

(1)Timer status bits are created automatically.

OPTIONAL:

WASH_DELAY.EN ⁽¹⁾		Spray Wash Delay Enabled
WASH_DELAY.TT ⁽¹⁾		Spray Wash Delay Timing

(1) Timer status bits are created automatically.

Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution "Change processor mode to Run?", select **Yes**.

HMI:

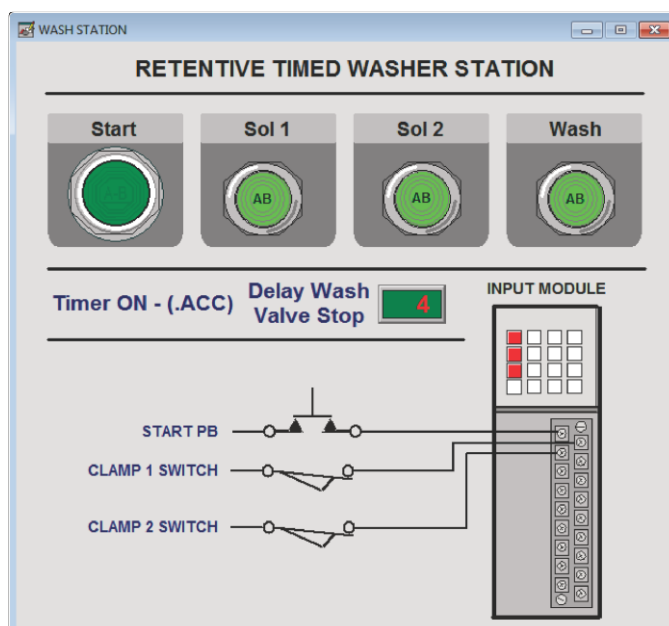
✓ Step ④ should be done before the washer times out.

① Operator Pressing and Holding the START pushbutton:

Placing the cursor over the **Start** pushbutton, left click and hold the mouse button.

If you do not hold the left mouse button the pushbutton is released.

② The two outputs for the solenoid valves for the clamps turn ON. The **Sol1** and **Sol2** indicators will illuminate green.



③ After the input switches for the **Clamp 1** and **Clamp2** are set, the spray **Wash** valve turns ON and the indicator will illuminate green.

④ Release the left mouse button. This releases the **Start** pushbutton and the solenoid valves turn OFF and the inputs for the clamps reset.

⑤ The timer's accumulated value is retained.

⑥ Operator Pressing and Holding the START pushbutton:

Placing the cursor over the **Start** pushbutton, left click and hold the mouse button.

If you do not hold the left mouse button the pushbutton is released.

⑦ After ten seconds the spray **Wash** turns OFF.

⑧ To turn open the clamps, release the left mouse button. This releases the **Start** pushbutton and the solenoid valves turn OFF and the inputs for the clamps reset.

Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: CONVEYOR TIMER

PROCESS FILE: CONVEYOR TIMER

OBJECTIVE: Use of TIMER ON and OFF instructions with cascading logic.

HMI: CONVEYOR-TIMER_HMI

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

Use delay to energize and delay to de-energize timers to create a delay Start/Stop Circuit.

When the Start pushbutton is depressed, the first conveyor output will be energized immediately, a second conveyor output will be energized after five (5) seconds.

The depressing of the Stop pushbutton will de-energize the first conveyor output after a five (5) second delay, and the second conveyor output will be de-energized after an additional five (5) second delay.

When creating the tag, the description MUST BE entered.

INPUT: (*Be sure to assign the correct Alias Address to the Tag*)

Alias Address	Controller Tag	Description
Local:3:I.Data.6	START	Momentary N.O. Pushbutton
Local:3:I.Data.7	STOP	Momentary N.C. Pushbutton

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	CONVEYOR_1	Conveyor Motor No. 1
Local:4:O.Data.1	CONVEYOR_2	Conveyor Motor No. 2

INTERNAL:

Param/Local Tag	Data Type	Description
RUN_CONTACT	BOOLEAN	Start Motors/Motors Running

TIMERS:

Param/Local Tag	Data Type	Description
CONV2_START_DELAY	TIMER	Delay Energize Timer for Conveyor No. 2
CONV2_START_DELAY.DN ⁽¹⁾		Conveyor No. 2 Start
CONV1_STOP_DELAY	TIMER	Delay De-energize Timer for Conveyor No. 1
CONV1_STOP_DELAY.DN ⁽¹⁾		Delay for Halting/Stop Conveyor No. 1
CONV2_STOP_DELAY	TIMER	Delay De-energize Timer for Conveyor No. 2
CONV2_STOP_DELAY.DN ⁽¹⁾		Delay for Halting/Stop Conveyor No. 2

(1)Timer status bits are created automatically.

OPTIONAL:

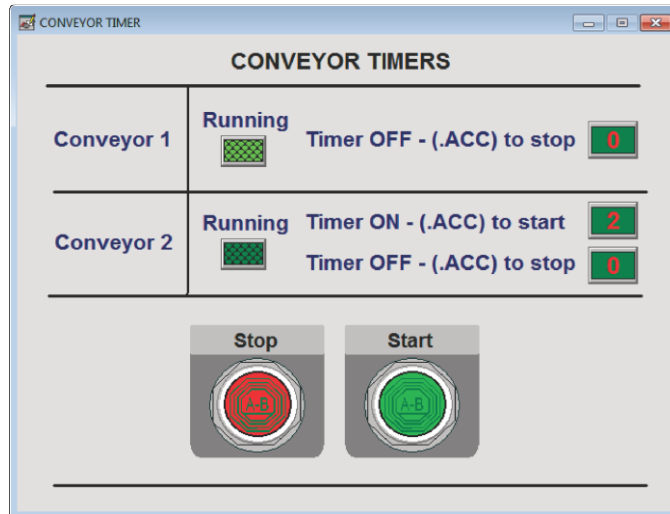
The ENABLE and TIMER TIMING status bits.

Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution "Change processor mode to Run?", select **Yes**.



HMI:

- ① Operator Pressing the START pushbutton:
Left click on the **Start** pushbutton.
- ② **Conveyor 1** should turn ON immediately.
- ③ After the time delay, **Conveyor 2** turns ON
- ④ Operator Pressing the STOP pushbutton:
Left click on the **Stop** pushbutton.
- ⑤ **Conveyor 1** turns OFF after the time delay.
- ⑥ **Conveyor 2** turns OFF after the time delay.

Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: PARTS DIVERTER

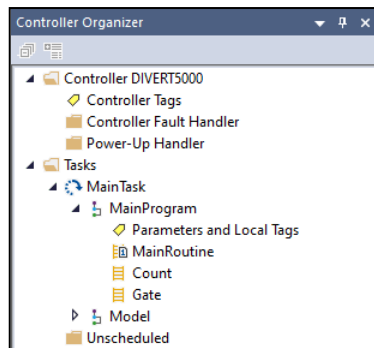
PROCESS FILE: DIVERT

OBJECTIVE: Use of counter to initiate a timed output.

HMI: DIVERT_HMI

SOFTWARE: Studio500, RSEmulate500, and FactoryTalk SE

Create a program that diverts one in five parts for a quality inspection. A START/STOP pushbutton station controls the parts conveyor. Parts are detected by a normally open proximity switch. When a count of 5 has been reached, a solenoid for a diverter gate is opened. The gate stays open for two seconds. The parts counter will reset to zero automatically after the fifth part. A manual reset pushbutton is provided to also return the counter's accumulated value to zero.



The program will use Subroutine for the Counter and Diverter Gate.

Subroutine Name: Count
Gate

[Appendix A - Adding a Subroutine](#)

[Appendix B - Adding a Jump to Subroutine Instruction](#)

When creating the tag, the description MUST BE entered.

INPUT: *(Be sure to assign the correct Alias Address, terminal numbers are not sequential, to the Tag)*

Alias Address	Controller Tag	Description
Local:3:I.Data.6	START	Momentary N.O. Pushbutton
Local:3:I.Data.5	RESET	Momentary N.O. Pushbutton
Local:3:I.Data.7	STOP	Momentary N.C. Pushbutton
Local:3:I.Data.0	PX1	Proximity Switch for Count

OUTPUTS:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	GATE	Diverter Gate
Local:4:O.Data.1	MOTOR	Parts Conveyor Motor

TIMER:

Param/Local Tag	Data Type	Description
GATE_TIMER	TIMER	Diverter Open Delay
GATE_TIMER.DN ⁽¹⁾	TIMER	Close Inspection Gate

(1) Timer status bits are created automatically

COUNTER:

Param/Local Tag	Data Type	Description
PART_COUNT	COUNTER	Parts Counter
PART_COUNT.DN ⁽¹⁾	COUNTER	Part Count Setpoint for Gate Open

(1) Counter status bits are created automatically

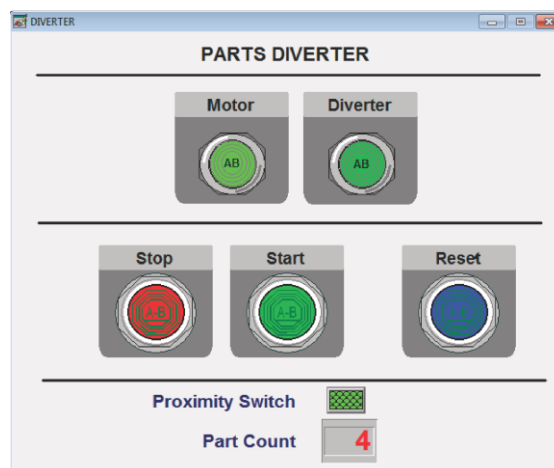
Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution "Change processor mode to Run?", select **Yes**.

HMI:



- ① Operator Pressing the START pushbutton:
Left click on the **Start** pushbutton.
- ② The conveyor **Motor** will turn ON.
- ③ As the conveyor runs, the **Proximity Switch** sensor will pulse OFF and ON.
- ④ When the **Part Count** is at 5, the **Diverter** turns ON.
- ⑤ After the time delay, the **Diverter** turns OFF.
- ⑥ Operator Pressing the STOP pushbutton:
Left click on the **Stop** pushbutton.

Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: SPOOL

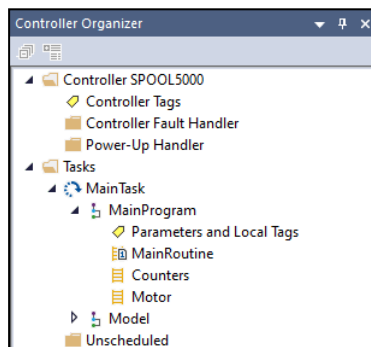
PROCESS FILE: SPOOL

OBJECTIVE: Input logic to control two counters and to have one counter control the spool motor output.

HMI: SPOOL_HMI

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

Write a program to implement the process outlined. A company which assembles electronic kits needs a counter to count and control the number of resistors placed into each kit. The controller must stop the take-up spool at a predetermined number of resistors (20). An operator will then cut the resistor strip and place it into the kit. Pressing and releasing the start pushbutton repeats the spooling/counting cycle.



The program will use Subroutines for the Counter and Motor.

Subroutine Names: Counters
Motor

[Appendix A - Adding a Subroutine](#)

[Appendix B - Adding a Jump to Subroutine Instruction](#)

The circuit operates as follows:

1. A START/STOP pushbutton station is used to turn the spool motor drive on and off manually. Pressing and releasing the START pushbutton begins the spooling/counting cycle
2. A thru-beam sensor counts the resistors as they pass by.
3. A Kit counter preset for 20 (the number of resistors in each kit) will automatically stop the take-up spool when the accumulated count reaches 20.
4. The operator cuts and removes the resistors from the take-up spool and then presses the Kit Counter RESET pushbutton.
5. A second counter is provided to count the grand total used.
6. Manual reset button is provided for the Grand Total Counter.

When creating the tag, the description MUST BE entered.

INPUT: (*Be sure to assign the correct Alias Address to the Tag*)

Alias Address	Controller Tag	Description
Local:3:I.Data.0	SENSOR_PE1	Thru-beam Sensor Detect Resistors
Local:3:I.Data.2	RESET_KIT_COUNTER	Kit Counter RESET, Momentary N. O. Pushbutton
Local:3:I.Data.3	RESET_TOTAL_COUNTER	Grand Total RESET, Momentary N. O. Pushbutton
Local:3:I.Data.6	START	Momentary N. O. Pushbutton
Local:3:I.Data.7	STOP	Momentary N. C. Pushbutton

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	SPOOL	Take-up Spool

COUNTERS:

Param/Local Tag	Data Type	Description
KIT_COUNTER	COUNTER	Kit Parts Counter
KIT_COUNTER.DN ⁽¹⁾		Kit Count Complete Stop Spool
TOTAL_COUNTER	COUNTER	Grand Total Counter Per Spool

⁽¹⁾ Counter status bits created automatically

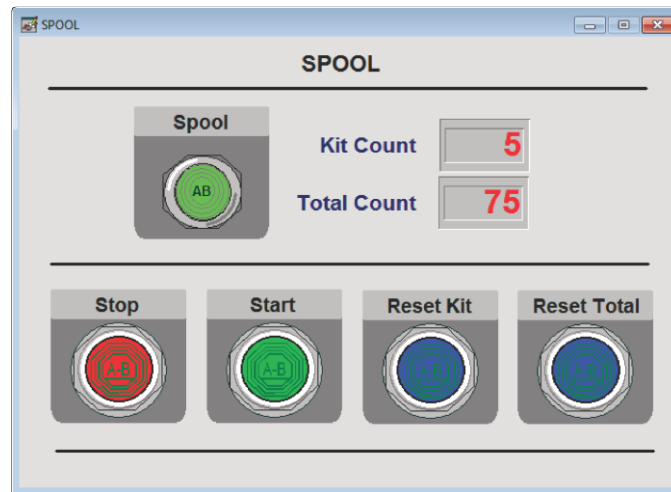
Studio 5000 Logix Emulate must be started before Downloading

Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution "Change processor mode to Run?", select **Yes**.

HMI:



- ① Operator Pressing the START pushbutton:
Left click on the **Start** pushbutton.
- ② The take-up **Spool** turns ON.
- ③ As resistors pass the photoelectric sensor, **Kit Count** and **Total Count** increments.
- ④ When the **Kit Count** is at 100 the **Spool** turns OFF.
- ⑤ Operator Pressing the RESET KIT pushbutton:
Left click on the **Reset Kit** pushbutton.
- ⑥ **Kit Count** returns to zero.
- ⑦ Operator Pressing the RESET TOTAL pushbutton:
Left click on the **Reset Total** pushbutton.
- ⑧ **Total Count** returns to zero.

Offline:

1. To return to Offline
 - a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
 - b. At the caution "Change processor mode to Remote Program?", select **Yes**.
 - c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: CONVEYOR TIMER with INDICATORS

PROCESS FILE: CONVEYOR INDICATORS

OBJECTIVE: Use both direct and indirect information to determine output action.

HMI: CONVEYOR-INDICATOR_HMI

SOFTWARE: Studio500, RSEmulate500, and FactoryTalk SE

A subroutine of the Conveyor Timer is included. You do not need to recreate the logic for the conveyor control. Add the **Tags** and **Ladder Logic subroutine** for the indicators.

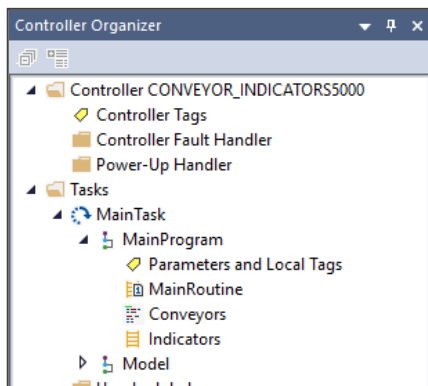
Modify the CONVEYOR TIMER lab with the following indicators for both conveyors: running, in a halting cycle, and stopped.

A RUNNING indicator will be energized as each of the conveyors is energized.

When the STOP pushbutton is pressed the RUNNING indicator for the first conveyor will be turned off. The HALTING indicator for the first conveyor will be turned on while the conveyor is timed delayed to be de-energized.

At the completion of the timing cycle and the first conveyor is off, a STOPPED indicator for the first conveyor will be turned on.

The second conveyor when Started or Stopped will have the sequence with the indicators, STOP → RUN → HALT → STOP.



The program will use a Subroutine for the indicators.

Subroutine Name: Indicators

[Appendix A - Adding a Subroutine](#)

[Appendix B - Adding a Jump to Subroutine Instruction](#)

When creating the tag, the description MUST BE entered.

OUTPUTS:

Alias Address	Controller Tag	Description
Local:4:O.Data.10	INDC_RUN_1	Conveyor No. 1 Running Indicator
Local:4:O.Data.11	INDC_RUN_2	Conveyor No. 2 Running Indicator
Local:4:O.Data.12	INDC_HALT_1	Conveyor No. 1 Halting Indicator
Local:4:O.Data.13	INDC_HALT_2	Conveyor No. 2 Halting Indicator

Local:4:O.Data.14	INDC_STOP_1	Conveyor No. 1 Stopped Indicator
Local:4:O.Data.15	INDC_STOP_2	Conveyor No. 2 Stopped Indicator

 Only one (1) status indicator per conveyor will be ON at a time.

	START Pushbutton Pressed		Both Running	STOP Pushbutton Pressed		Both Stopped
INDC_STOP1						
INDC_HALT1						
INDC_RUN1						
INDC_STOP2						
INDC_HALT2						
INDC_RUN2						

Subroutine: **Indicators**

Create the Subroutine, **Indicators**, for the logic to control the conveyors status indicators.

MainProgram:

Add the Subroutine call to the Function Indicators.

Studio 5000 Logix Emulate must be started before Downloading

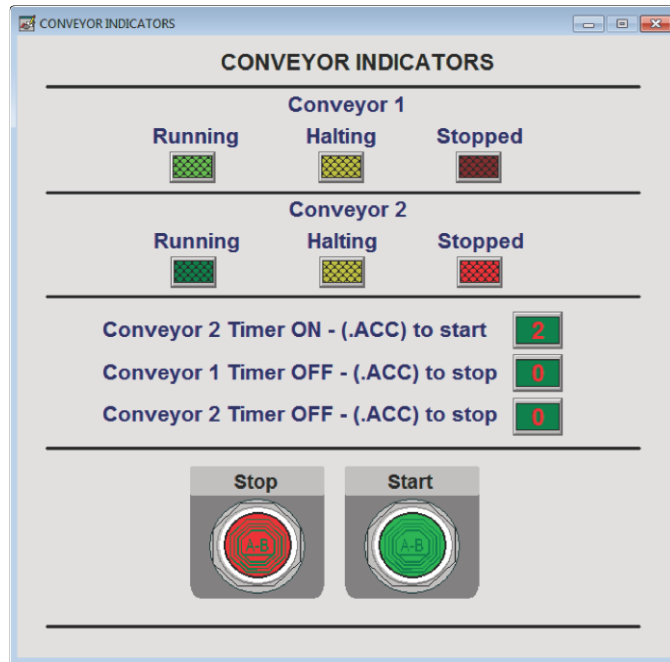
Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution "Change processor mode to Run?", select **Yes**.

continued

HMI:



- ① Operator Pressing the START pushbutton:
Left click on the **Start** pushbutton.
- ② **Conveyor 1** should turn ON immediately, the **Stopped** indicator is OFF and **Running** indicator is ON.
- ③ After the time delay, **Conveyor 2** turns ON and **Running** indicator is ON, the **Stopped** indicator is OFF and **Running** indicator is ON.
- ④ Operator Pressing the STOP pushbutton:
Left click on the **Stop** pushbutton.
- ⑤ The Conveyor 1 **Running** indicator turns OFF and the **Halting** indicator is ON.
- ⑥ After the time delay, the Conveyor 1 **Halting** indicator turns OFF and the **Stopped** indicator is ON.
- ⑦ The Conveyor 2 **Running** indicator turns OFF and the **Halting** indicator is ON.
- ⑧ After the time delay, the Conveyor 2 **Halting** indicator turns OFF and the **Stopped** indicator is ON.

continue

Offline:

To return to Offline

- a. At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- b. At the caution "Change processor mode to Remote Program?", select **Yes**.
- c. At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

LAB: MOVE

PROCESS FILE: MOVE

OBJECTIVE: Use MOVE instructions to change a Timer's Preset value

SOFTWARE: Studio 5000, Studio 5000 Logix Emulate, and FactoryTalk SE

Use switches with MOVE instruction to change a Timer's Preset value based on the value in the MOVE instruction.

A START-STOP pushbutton station is used to turn on an OUTPUT for the duration of the timer. When the timing is complete the timer's accumulated value returns to zero. Pressing STOP will return the timer's accumulated value return to zero.

When creating the tag, the description MUST BE entered.

INPUT:

Alias Address	Controller Tag	Description
Local:3:I.Data.0	STOP	Momentary NC Pushbutton Stop
Local:3:I.Data.1	START	Momentary NO Pushbutton Start
Local:3:I.Data.2	SW_PRESET1	Momentary NO Pushbutton Preset 1
Local:3:I.Data.3	SW_PRESET2	Momentary NO Pushbutton Preset 2
Local:3:I.Data.4	SW_PRESET3	Momentary NO Pushbutton Preset 3

OUTPUT:

Alias Address	Controller Tag	Description
Local:4:O.Data.0	OUTPUT	Delayed Output

INTERNAL: Note: the internal is a Program Tag.

Param/Local Tag	Data Type	Description
DWELL	TIMER	Delay Timer for Output

I/O CONFIGURATION: (Processor and I/O has been configured)

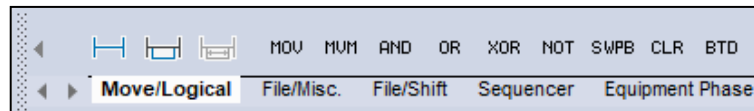
Slot 2 - Studio Emulate 5000 Controller

Slot 3- 1756-IB16

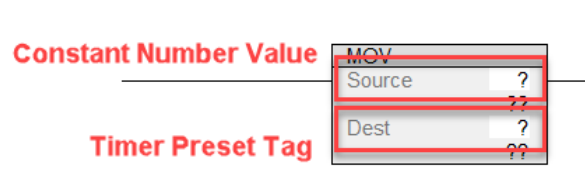
Slot 4 - 1756-OB16E

MOVE Instruction:

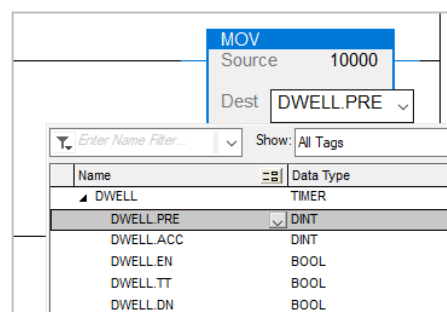
The Move (**MOV**) instruction is on the Instruction Palette, **Move/Logical** tab.



When entering the **Source** for the Move instruction, type the Constant number values of 10000, 15000, and 20000 for the appropriate Move instruction. The **Destination** is the Timer's Preset Tag.

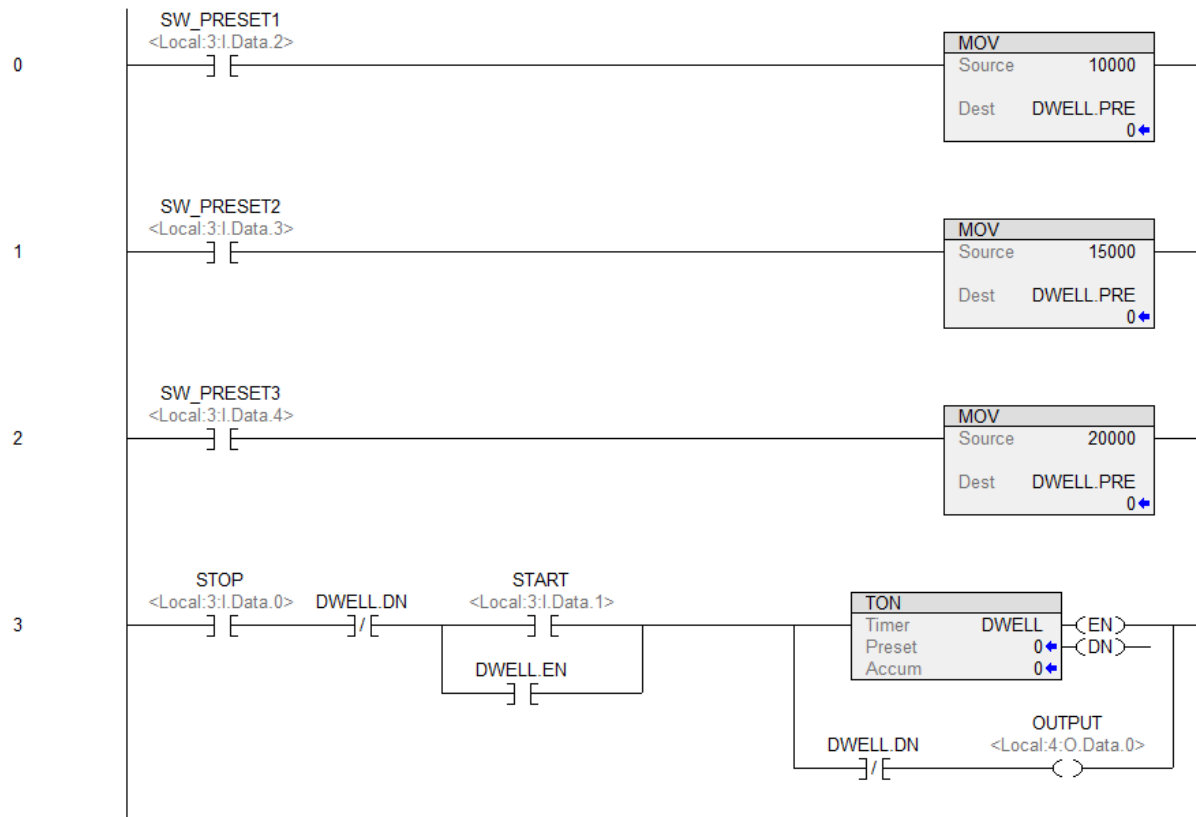


To select the Timer's Preset, use the dropdown and expand the Tag Name for the .PRE tag.



SEE NEXT PAGE FOR PROGRAM LOGIC.

Program Logic:



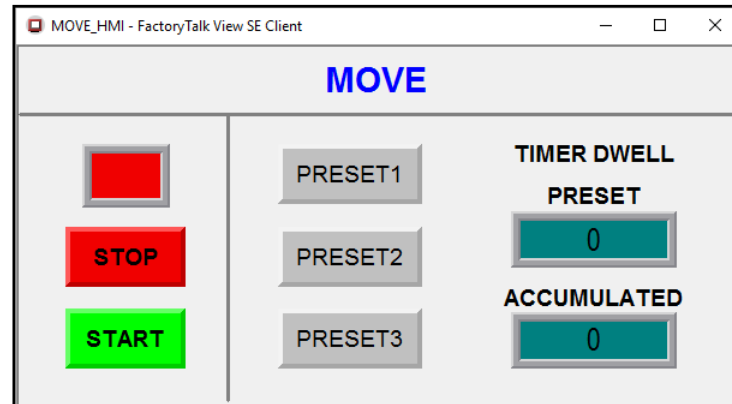
Download:

Download the project and Run.

- At the ONLINE BAR go from **OFFLINE** to **Download**.
- At the caution, select the **Download** button.
- At the ONLINE BAR go from **Rem Prog** to **Run**.
- At the caution "Change processor mode to Run?", select **Yes**.

SEE NEXT PAGE FOR HMI

HMI:



① Press either PRESET1, PRESET2, or PRESET3 button.

② Press the START button.

The OUTPUT indicator is On and the timer is elapsing.

When the timer times out, the OUTPUT turns OFF and the Accumulated value returns to zero.

③ Repeat the PRESET select with a different button and START the Timer.

④ START the timer, then press the STOP button.

The OUTPUT indicator turns OFF and the Accumulated value returns to zero.

Offline:

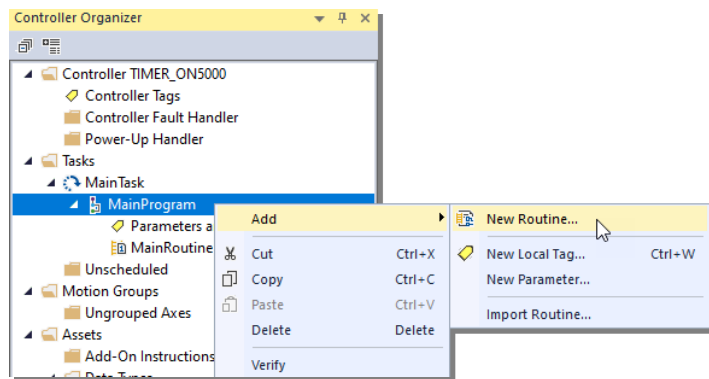
To return to Offline

- At the ONLINE BAR go from **Rem Run** to **Program Mode**.
- At the caution "Change processor mode to Remote Program?", select **Yes**.
- At the ONLINE BAR go from **Rem Prog** to **Go Offline**.

APPENDIX A - ADDING A SUBROUTINE

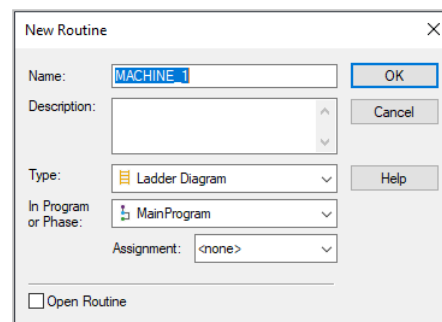
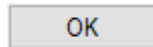
In the **Controller Organizer**, right click on **MainProgram**.

Select **Add** then **New Routine...**

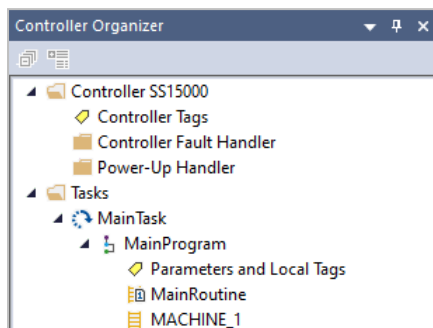


Type in the Subroutine in the **Name:** field.

Select the **OK** button.



Routine is added to the project.



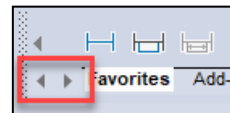
[Return to Parts Diverter](#)

[Return To Spool](#)

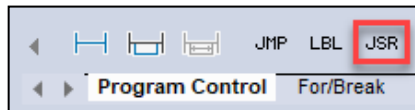
[Return Conveyor Timer with Indicators](#)

APPENDIX B - ADDING A JUMP TO SUBROUTINE INSTRUCTION

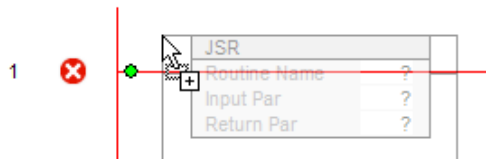
Use the Next Tab button to select the **Program Control** tab.



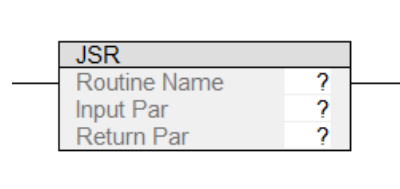
In the **Program Control** tab, select the **JSR** instruction by left click and hold.



Drag the instruction to a green marker on the required rung.

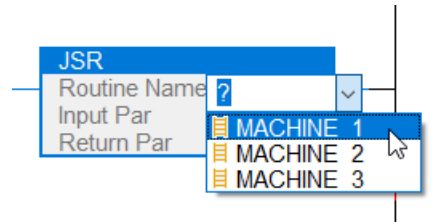


The instruction is added to the right side of the rung.



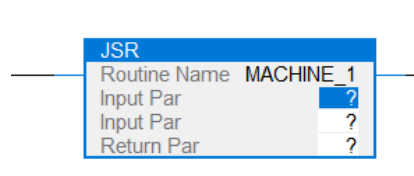
Double left click on the **Routine Name** field.

Left click on the down arrowhead and select the required SubRoutine name.

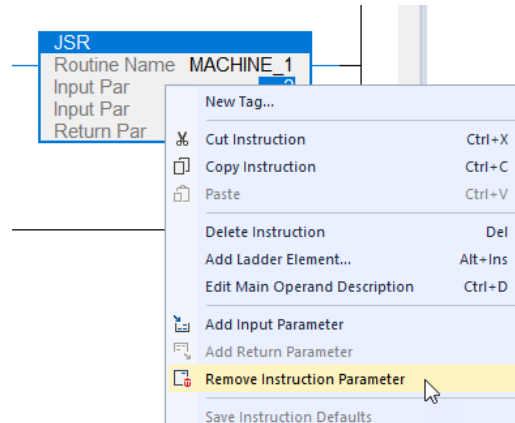


Parameters will not be used in these labs.

Right click on the parameter field.

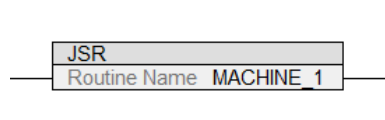


Select **Remove Instruction Parameter**.



Repeat the above two steps for the remaining Parameters.

Completed Jump to Subroutine.



[Return to Parts Diverter](#)

[Return To Spool](#)

[Return Conveyor Timer with Indicators](#)